Project Geminae

Change 6 — Missingness Indicator for Injection Pressure (P)

Example validation: `dowhy_mediation_analysis_30d_20260402_165115.csv` | *30-day lookback × 20 radii × multiple estimators*

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1. Change Summary

Change 6 replaces **median imputation** of the mediator pressure column **P** (e.g. `Injection Pressure Average PSIG`) with a two-part handling intended to separate *structural* missingness from *numeric* pressure:

- **Zero-fill P:** missing values in P are set to `0` instead of the column median.
- **Binary `P_missing`:** a 0/1 indicator is set to 1 on rows where P was missing *before* the fill; this variable enters the causal DAG as an additional measured covariate (with edges from G1, G2, W and to S, consistent with informative missingness or reporting gaps).

The prior approach (median-fill P together with G1/G2) implicitly treated all missing pressures as exchangeable
with "typical" active-well pressures. Zero-fill encodes "no reported pressure" as a low-pressure state while `P_missing` allows the model to adjust for whether that state is likely **idle / non-reporting** versus a **data gap** — a distinction that requires sponsor / domain input to interpret causally.

2. Issue Addressed (Pre–Change 6)

Median imputation of P confounds scale and missingness. Filling NaN pressure with the median of *observed* pressures assigns high typical operating pressures to rows that may represent non-injecting wells, reporting lapses, or true zeros. That can bias path *a* (W → P), path *b* (P → S | W), and mediation decomposition. Change 6 does not claim to identify *why* P is missing; it adds a transparent encoding (0 + indicator) and explicit adjustment so estimates can shift when missingness is prevalent.

3. Methodology

Preprocessing (per analysis script, at read time): After column auto-detection, the code computes `P_missing = P.isna()` (as float 0/1), then `P = P.fillna(0)`. Geology covariates G1 and G2 continue to be median-imputed where applicable; **P is not median-imputed**.

DAG extension: Graph nodes include `P_missing` with directed edges (G1 → P_missing), (G2 → P_missing), (W → P_missing), (P_missing → S), in addition to the

existing W → P → S / W → S structure. DoWhy's backdoor sets update accordingly; manual OLS paths and nonlinear first stages include `P_missing` alongside G1, G2 (and `well_count` at event level).

Event-level aggregation (`dowhy_ci_aggregated.py`, `dowhy_simple_all_aggregate.py`): Row-level `P_missing` is computed before `groupby(EventID)`; after zero-fill, P is aggregated as before (median across wells in radius). `P_missing` is aggregated as the **mean** of the well-level indicators, i.e. the **fraction of linked wells in that radius with missing P** for that event.

Output diagnostic: `dowhy_ci.py` and `dowhy_ci_aggregated.py` add a column `pressure_missing_pct` to the results table: `100 × mean(P_missing)` for that radius file (or event-level dataset), so dashboards and briefs can track how much missingness the run actually saw.

Interaction with Change 3 (log transform): `P_missing` is not log-transformed. After zero-fill, `log1p(P)` maps true zeros to 0; rows that were missing are 0 on P and 1 on `P_missing`, preserving separability in the design matrix.

4. Validation Run — Diagnostic Column (`pressure_missing_pct`)

Observation	Interpretation
`pressure_missing_pct = 0` for all radii in sample CSV	The mediator column matched by `["pressure","psig"]` contained **no pandas NaN** values in `event_well_links_with_faults_30d_*km.csv` for that run. Change 6 still executes (zero-fill and DAG are active); the indicator is simply all zeros.
Non-zero `pressure_missing_pct` in future runs	Expected when raw SWD/reporting leaves pressure blank for some rows. Effect estimates may shift relative to the all-observed-pressure case; review path *a* / *b* and % mediated alongside this diagnostic.

To verify missingness in a fault file: `df[pressure_col].isna().mean()` before any fill.
Empty strings or numeric placeholders coded as 0 are not counted as missing unless parsed as NaN.

5. Key Findings and Interpretation

- **Change 6 is a modelling hygiene fix, not a guarantee of larger causal effects.** When missingness is rare, posterior/point estimates may be nearly unchanged; the value is correct handling when missingness is non-negligible.
- **`pressure_missing_pct` is a reporting diagnostic.** It answers "what fraction of rows had missing P before imputation?" not "is the model valid?" Sponsor review should tie high values to operational reporting rules (idle wells, optional fields, etc.).
- **Zero vs median is a substantive choice.** Zero aligns with "no gauge reading / no pressure"; median aligns with "typical operating well." The indicator partially disentangles these only if missingness is informative and correctly encoded as NaN upstream.

- **Legacy OLS batch scripts unchanged in scope.** `dowhy_simple_all.py` and `dowhy_simple_all_aggregate.py` implement Change 6 but still read only `30 day lookback window` fault filenames by design; multi-window production remains centered on `dowhy_ci.py --lookback-days N` (Change 5).

*Sponsor / domain input remains required: is missing P primarily **informative** (e.g. well idle) or **non-informative** (reporting gap)? The DAG treats `P_missing` as a measured covariate affecting S; if the true data-generating process differs, sensitivity analysis (e.g. exclude rows with `P_missing = 1`, or alternative fills) should be documented alongside production results.*

6. Files Changed

File	Change
`dowhy_ci.py`	`P_missing` before zero-fill on P; median-fill only G1/G2; extended DAG and all mediation / refutation / predictive design matrices; `pressure_missing_pct` in per-radius output rows
`dowhy_ci_aggregated.py`	Same pattern at well level before event aggregation; `P_missing` mean per event; extended DAG and OLS / VIF / predictive blocks; `pressure_missing_pct` in summary output
`dowhy_simple_all.py`	Well-level OLS mediation path: `P_missing`, zero-fill P, extended DAG, predictive features include `P_missing`
`dowhy_simple_all_aggregate.py`	Event-level OLS path: indicator + zero-fill before `groupby`; `P_missing` as fraction; extended DAG and predictive features

ETL scripts (`filter_active_wells_before_events.py`, etc.) are unchanged; missingness is handled at causal-model ingest unless a future change persists `P_missing` in exported CSVs.

7. Next Steps

Production alignment	Re-run `python dowhy_ci.py --lookback-days 90` (or canonical N) with Change 6 on the same fault CSVs used for sponsor-facing tables; archive CSV with non-zero `pressure_missing_pct` if subsamples show missing P.
Refutation columns	If `placebo_effect`, `subset_effect`, or `dowhy_original_effect` are empty in output, inspect console logs for refuter failures; orthogonal to Change 6 but affects completeness of the mediation export.
Change 7	Two-stage hurdle model (occurrence vs magnitude) as planned; builds on stable window (Change 5) and pressure handling (Change 6).